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Dr. Jon R. Fox

Applied Physicist with 25 years of post-doctoral experience spanning MEMS device design and fabrication, model-based systems engineering, and signal analysis and processing, including multiphysics modeling, hardware test, and digital engineering environments. Programmer and data wrangler for microcontrollers, embedded systems, automated test hardware, workstations, and cloud-based servers.

Summary of Skills

- Ph.D. in experimental condensed-matter physics.
- Division Chief Scientist, Architecture and System Development Division of the Applied Systems Laboratory (GTRI).
- Model-Based Systems Engineering with OMG SysML (v1 and v2), MagicDraw/Cameo Enterprise Architecture, and FACE conformance tooling.
- Open-source developer: author of SysML v2 tooling in Python (sysmlpy, uml2py, sysml_canvas) with international collaborators.
- Innovator and funded author on multiple research, development, and commercialization efforts.
- Capable technical writer and presenter.
- 25+ years of data acquisition and analysis using Python, C/C++, FORTRAN (77/90/95), Go, Groovy, Java, LabVIEW.
- Adept user (and admin) of Unix (Linux, Solaris) and Windows environments, office software suites, and SQL and Graph databases.

Experience and Accomplishments

January 2024 - Present

Division Chief Scientist, Architecture and System Development Division, Applied Systems Laboratory, Georgia Tech Research Institute, Huntsville, AL.

- Technical direction, review, and mentoring across division programs in model-based systems engineering, architecture, and system development.

September 2019 - Present

Principal Research Scientist, Architecture and System Development Division, Applied Systems Laboratory, Georgia Tech Research Institute, Huntsville, AL.

- Model Based Systems Engineering technical analyses for US Army Aviation Future Vertical Lift systems and software architecture.
- GTRI Technical Lead for Future Long-Range Assault Aircraft (FLRAA) architectural risk reduction efforts, utilizing the Future Vertical Lift Architectural Framework (FAF) to enable digital engineering of variant rotorcraft architectures and incorporating the work of multiple Tier 1, 2, and 3 corporate architectural partners.

- Division Chief Scientist of the Architecture and System Development Division, Applied Systems Laboratory, providing technical direction, review, and mentoring across division programs.
- Participating Member of the Architecture Collaboration Working Group (ACWG) and participant in multiple sub-groups.
- Jython and Groovy scripting for data extraction and addition into Cameo Enterprise Architecture.
- FACE data modeling with the FAME Awesum tool by Tucson Embedded.
- Open-source development of SysML v2 tooling: clean-room Python implementations of the UML 2.5.1 metamodel and SysML v1-to-v2 transformation (uml2py), a SysML v2 reference API and conformance toolchain (sysmlpy, 123/123 OMG XPECT conformance), an interactive diagramming canvas (sysml_canvas), and a SysML v2 model of the AMS Government Reference Architecture.
- Clean-room Python ports of the FACE Conformance Test Suite data tooling, validated byte-for-byte against the reference Java tools.

March 2010 - August 2019

System and Software Architect, Project Manager, Applied Scientist, General Atomics Electromagnetic Systems Group, Huntsville, AL.

- Modeling of arresting gear system, electromagnetic railgun system, and laser effect system using the OMG Systems Modeling Language (SysML).
- Modeling included: use cases, requirements, block definition and internal block diagrams, requirements diagrams, and state machine diagrams using No Magic's MagicDraw. Additional MagicDraw functionality added using Jython and Swing GUI.
- Integration of SysML architectural models with Windchill product lifecycle management tool.
- Lead modeling and simulation of proposed linear displacement transducer systems for requirements verification and design downselect.
- Analysis and modeling of pressure data for real-time health prognostication.
- Development of automated generation of software design documentation from SysML models using Velocity Templating Language (VTL) and Ruby scripting.
- Optimization of hypersonic flight trajectories with NASA's OTIS tool.
- Information and network security analysis with CompTIA Security+ certification.
- Program and functional management role for remaining SETA work with AMRDEC/CCDC AMC until Mar. 2018.
- Provided science and engineering support to the Charles Bowden Weapons Sciences Laboratory of AMRDEC-WDI, leading in the development of MEMS and nanomaterials for nano-composite piezoelectric microphones, surface micro-machined aluminum micro-mirrors, low-pressure microscale sub-die pressure transducers for hardware health monitoring, surface acoustic health monitoring of die and printed wiring boards, magnetic micro-actuators for missile CAS, and MEMS vibrational energy health transducers for missile health prognostication.
- Functional manager of employees supporting work at the Weapons Sciences Laboratory.

July 2007 - March 2010

Thin Film Process Engineer, Gulton, Inc., South Plainfield, NJ.

- Successfully developed thin film thermal print head (TPH) product lines for a small company with no corporate thin film manufacturing experience.

- Process control and product management of existing thick film TPHs from screening to final assembly.
- Selection, installation, maintenance of production physical vapor deposition equipment.
- Developed dry and wet etch processes for metals, resistor materials, and passivation dielectrics from prototype to production.
- Programmed automated test equipment for resistor testing and trimming at the die level and the packaged part level.
- Developed and implemented web application servers for SPC data collection and analysis across the factory floor.

April 2000-July 2007

Principal Scientist, Research Support Instruments, Princeton, NJ.

- Conceived and managed numerous technology development projects in micro-fabricated structures.
- Appointed Visiting Professional Technical Staff Member II in the Department of Mechanical and Aerospace Engineering of Princeton University (2000–2006).
- Industrial Partner with Princeton Institute of the Science and Technology of Materials.
- Awarded multiple research grants towards RSI micro-machining efforts from the SBIR/STTR program with AFRL, ONR, NIST, & NASA.
- Principal Investigator/Project Manager, NASA Small Business Innovation Research (SBIR) Grant Phase I “An inexpensive, versatile micro-machined mass spectrometer.”
- Principal Investigator/Project Manager, Air Force SBIR Grant Phase I & II, “A micro-machined energy-mass spectrograph for micro-satellite applications.”
- Principal Investigator/Project Manager, NIST SBIR Grant Phase I, “A Micro-Electroformed Cross Capacitor.”
- Principal Investigator/Project Manager, ONR STTR Grant Phase I, “Passive MEMS tagging of Personnel.”
- Modeled the performance of micro-fabricated ion optics using Numeric Python and OpenDX.

1993-2000

Research Assistant, Penn State Physics Department, University Park, PA.

- Developed skills in many analytical techniques for materials and surfaces: including Raman Spectroscopy, Photoluminescence, Electron Energy Loss Spectroscopy (HREELS), Photoelectron Emission Spectroscopies (UPS, XPS, AES), LEED and RHEED, and powder neutron diffraction.
- Studied a wide range of novel materials: fullerenes; Si, Ge, and C amorphous and crystalline nanoparticles; ferromagnetic oxides; and II-VI epitaxial films.
- Developed a wide range of high and ultra-high vacuum materials deposition skills including rf & dc magnetron sputtering, e-beam evaporation, and pulsed laser deposition.
- Authored peer reviewed scientific papers and funding reports for the National Science Foundation and Department of Energy.

Education

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| <i>Ph. D. Physics</i> | The Pennsylvania State University, University Park, PA, May 2000. Thesis: “ <i>in situ</i> studies of the electronic and vibrational properties of thin films and novel materials”, Advisor: the late Professor Jeff Lannin. Thesis completed under the supervision of committee chair Professor Robert Collins. |
| <i>B.S. Physics</i> | <i>Magna cum laude</i> . University of Missouri-Rolla, Rolla, MO, 1993. Minor in mathematics. |

Current Memberships

- IEEE/ Computer Society
- the International Council on Systems Engineering (INCOSE)

Current Certifications

- INCOSE Certified Systems Engineering Professional (CSEP), March 2025

Open Source & Publications

- Author and maintainer of the sysmlpy Python toolchain for the OMG SysML v2 standard (parser, semantic analyzer, API services), with international collaborators and OMG conformance testing. github.com/mycroft
- Curator of the *awesome-sysml* list of SysML v2 language tools.

- [1] Stephen B. Horowitz, Michael S. Allen, Jon R. Fox, Jean P. Cortes, Laura Barkett, Adam D. Mathias, Corey Hernandez, Adam C. Martin, Mohan Sanghadasa, Stephen Marotta “A low frequency MEMS vibration sensor for low power missile health monitoring.” 2013 IEEE Aerospace Conference, Big Sky, MT, March 2013.
- [2] Jon R. Fox, Adam D. Mathias, Jean Cortes, Michael S. Allen, Stephen B. Horowitz, Mark G. Temmen, and Mohan Sanghadasa. “Fabrication and characterization of ultra-fast electrostatically-actuated surface micro-machined aluminum mirrors.” American Vacuum Society 59th Annual International Symposium and Exhibition held in Tampa, Florida on October 30, 2012.
- [3] J.R. Fox, S.B. Horowitz, J.P. Cortes, M.S. Allen, A.D. Mathias, L.A. Barkett, and M. Sanghadasa. “Fabrication and Testing of Suspended Piezoelectric Nanocomposite Membranes.” American Vacuum Society 59th Annual International Symposium and Exhibition held in Tampa, Florida on October 29, 2012.
- [4] Stephen Horowitz, Jon Fox, Dustin Mathias, Jean Cortes, Michael Allen, Laura Barkett, Mohan Sanghadasa. “Miniaturized Nanocomposite Piezoelectric Microphones for UAS Applications.” 2012 Joint Meeting of the Military Sensing Symposia (MSS) held in Washington, DC on October 22-25, 2012.
- [5] Stephen Horowitz, Dustin Mathias, Jon Fox, JP Cortes, Mohan Sanghadasa, Paul Ashley. “Effects of Scaling and Geometry on the Performance of Piezoelectric Microphones.” Sensors and Actuators, 185: 24-32, October 2012.
- [6] Adam D Mathias, Jon R Fox, Jean P Cortes, Stephen B Horowitz, Mohan Sanghadasa, Paul Ashley. “Recent progress toward a nanostructured piezoelectric microphone.” The Journal of the Acoustical Society of America Volume 130 (4) 2427, April 2011.
- [7] Adam D Mathias, Jon R Fox, Stephen B Horowitz, Mark G Temmen. A High Speed Electrostatically Actuated Al-MEMS Micromirror Array. Infotech@ Aerospace 2011, March 29, 2011.
- [8] M. M. Salamon, J. R. Fox, D. J. Sullivan, J. F. Kline, “Fabrication of a novel suspended microactuator MEMs device.”, Society of Manufacturing Engineers MicroManufacturing Conference, Minneapolis, Minnesota, Mar. 2009.
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- [10] Kline, J., J.R. Fox, V. DiPietro, and J. Brandenburg. A Review of the Compact Micro-Torus eXperiment. *4th Symposium on the Current Trends in International Fusion Research: A Review*. Washington, D.C. 2001.
- [11] John F. Kline, Jon R. Fox, Vincent DiPietro, John Vandenburg, Characterization of the Compact Micro Torus eXperiment. The 27th IEEE International Conference on Plasma Science. New Orleans, LA. June 6, 2000.
- [12] Kline, J., J.R. Fox, V. DiPietro, and J. Brandenburg. Propulsion Applications of the Compact Micro-Torus eXperiment. *NASA Fusion Propulsion Workshop*. 2000.
- [13] A. Sirenko, J. R. Fox, I. A. Akimov, X.X. Xi, S. Rouvimov, and Z. Lilienthal-Weber. *In situ* Raman scattering studies of amorphous and crystalline states in Si nanoparticles. *Solid State Commun.*, 113:553, 2000.
- [14] X. X. Xi, H. C. Li, W. D. Si, A. A. Sirenko, I. A. Akimov, J. R. Fox, A. M. Clark, J. H. Hao. Oxide thin films for tunable microwave devices. *Journal of Electroceramics*, 4 (2-3): 393-405, 2000.
- [15] J. R. Fox, I.A. Akimov, X.X. Xi, and A. A. Sirenko. In situ studies of electronic and vibrational properties of Si nanoparticles. In L.T. Canham, *et al.* , editors, *Advances in Microcrystalline and Nanocrystalline Semiconductors*, volume 536, page 287. Materials Research Society, 1998.
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- [17] G.P. Lopinski, J.R. Fox, J.S. Lannin, F.S. Flack, and N. Samarth. Reconstruction induced changes in the electronic states of ZnSe(100). *Surf. Sci. Lett.*, 355(1/3):L355, 1996.
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- [19] G. P. Lopinski, M.G. Mitch, J. R. Fox, and J. S. Lannin. Phase transition in RbC₆₀: Vibrational and electronic properties. *Phys. Rev. B., Rapid Commun.*, 50(21): 16098-16101, 1994.